

nuclear notation

Notation is A_ZX

where:

X = chemical symbol of element

A = nucleon number (no. of protons + no. of neutrons)

Z = proton number

$A = Z + N$ if and only if N = the neutron number

Equations for decay processes **must balance** in terms of both A and Z .

	Decay name			
	Alpha (α)	Beta-minus (β^-)	Beta-plus (β^+)	Gamma (γ)
Nature of emission	helium (${}^4_2\text{He}^{2+}$) nucleus $2n + 2p$	electron emitted from the nucleus e^-	positron (anti-electron) emitted from the nucleus e^+	electromagnetic radiation (photon) of large energy
Origin	overall change in neutron : proton ratio	neutron changes to proton	proton changes to neutron	removal of excess energy from nucleus nucleus moves to de-excited state
Ionizing power of radiation	strong	medium	quickly annihilated by nearby electrons	weak
Penetration	few cm of gas or few mm of paper	few cm of aluminium		many metres of lead or concrete
Notes	limited number of final energy states for a particular decay	broad energy spectrum of emitted electrons electron antineutrino is also emitted and carries off some energy	broad energy spectrum of emitted positrons electron neutrino is also emitted and carries off some energy	discrete energies linked to nuclear energy levels
Decay equation	${}^A_ZX \rightarrow {}^{A-4}_{Z-2}Y + {}^4_2\alpha^{2+}$	${}^A_ZX \rightarrow {}^A_{Z+1}Y + {}^0_{-1}\beta^- + \bar{\nu}_e$	${}^A_ZX \rightarrow {}^A_{Z-1}Y + {}^0_{+1}\beta^+ + \nu_e$	${}^A_ZX^* \rightarrow {}^A_ZX + \gamma$